

Population Risk and Bayes Estimator - brief recap

- **Setting:** data $\{(x_i, y_i)\}_i$ i.i.d. and drawn from a distribution p defined over $\mathcal{X} \times \mathcal{Y}$. In particular, p has a marginal distribution p_x over $\mathcal{X}$ and conditional distribution $p(\cdot | x)$ over $\mathcal{Y}$, i.e. $p(x, y) = p_x(x)p(y | x)$.
- **Population risk** for a predictor $f : \mathcal{X} \rightarrow \mathcal{Y}$ with respect to a loss function $\ell(y, z)$:

$$R(f) = \mathbb{E}_{(x,y) \sim p}[\ell(y, f(x))] = \mathbb{E}_{x \sim p_x}[\mathbb{E}[\ell(y, f(x)) | x]] = \mathbb{E}_{x \sim p_x}[r(f(x) | x)],$$

where we have defined the *conditional risk* $r(z | x) = \mathbb{E}[\ell(y, z) | x]$.

- **Bayes predictor:** a predictor f_* minimizing the population risk R . This implies that, for all $x \in \mathcal{X}$, the predictor evaluated at x is a minimizer of the conditional risk:

$$f_*(x) \in \arg \min_{z \in \mathcal{Y}} [r(z | x)].$$

The following exercise statements are partly selected from the book *Learning Theory from First Principles* by Francis Bach.

Exercise 1. ℓ_2 vs ℓ_1 loss in regression

Consider the following losses

$$\ell_2(y, z) = (y - z)^2, \quad \ell_1(y, z) = |y - z|.$$

and the associated empirical risks written respectively

$$\hat{R}_n^{(2)}(z) = \frac{1}{n} \sum_{i=1}^n \ell_2(y_i, z), \quad \hat{R}_n^{(1)}(z) = \frac{1}{n} \sum_{i=1}^n \ell_1(y_i, z).$$

for the ordered values $y_1 \leq y_2 \leq \dots \leq y_n$.

1. Show that the value z_2^* minimizing $\hat{R}_n^{(2)}(z)$ is the mean of the y_i 's, ie

$$z_2^* = \frac{1}{n} \sum_{i=1}^n y_i.$$

2. Let $k \in \{1, \dots, n-1\}$. Show that $z \mapsto \hat{R}_n^{(1)}(z)$ is differentiable in $]y_k, y_{k+1}[$ and compute its derivative within this interval. What would be its minimum within this interval?
3. Show that the value z_1^* minimizing $\hat{R}_n^{(1)}(z)$ is the median of the y_i 's, ie

$$z_1^* = \begin{cases} y_{(n+1)/2} & \text{if } n \text{ is odd,} \\ \frac{1}{2} (y_{n/2} + y_{n/2+1}) & \text{if } n \text{ is even} \end{cases}$$

4. What happens to z_1^* and z_2^* if we change y_n to $y_n + \Delta$ for some $\Delta \in \mathbb{R}$? Which of the two losses is more robust to outliers?

Exercise 2. Binary classification, weighted loss

Consider binary classification with $\mathcal{Y} = \{-1, 1\}$ with the loss function $\ell(-1, -1) = \ell(1, 1) = 0$ and $\ell(-1, 1) = c_- > 0$ (cost of a false positive), $\ell(1, -1) = c_+ > 0$ (cost of a false negative).

1. Write the conditional risks $r(-1 | x)$ and $r(1 | x)$ in terms of $p(1 | x)$ and $p(-1 | x)$.
2. Compare these two risks and determine the optimal prediction as a function of $p(1 | x)$.
3. Use $\mathbb{E}[y | x] = 2p(1 | x) - 1$ to express the Bayes predictor as a function of $\mathbb{E}[y | x]$.

Solution: Normalization of the conditional distribution

$$\sum_{y=\pm 1} p(y | x) = 1 \implies p(-1 | x) + p(1 | x) = 1. \quad (1)$$

Consider the conditional risk

$$\begin{aligned} r(z | x) &= \mathbb{E}[\ell(y, z) | x] = \sum_{y=-1,1} \ell(y, z)p(y | x) \\ &= \ell(1, z)p(1 | x) + \ell(-1, z)p(-1 | x) \\ &= \begin{cases} c_+p(1 | x), & \text{for } z = -1, \\ c_-p(-1 | x) = c_-(1 - p(1 | x)), & \text{for } z = 1. \end{cases} \end{aligned}$$

Therefore, $z = -1$ minimizes the conditional risk if $c_+p(1 | x) < c_-(1 - p(1 | x))$, hence $p(1 | x) < \frac{c_-}{c_+ + c_-}$; otherwise the minimizer is $z = 1$. Then, the Bayes Predictor is

$$f_\star(x) = \begin{cases} -1, & \text{if } p(1 | x) < \frac{c_-}{c_- + c_+}, \\ 1, & \text{if } p(1 | x) > \frac{c_-}{c_- + c_+}. \end{cases}$$

In order to express it as a function of the conditional expectation $\mathbb{E}[y | x]$, we recall its definition:

$$\mathbb{E}[y | x] = \sum_{y=-1,1} yp(y | x) = p(1 | x) - p(-1 | x) = 2p(1 | x) - 1 \implies p(1 | x) = \frac{1}{2}(\mathbb{E}[y | x] + 1).$$

It follows that, defining $\Delta = (c_- - c_+)/ (c_- + c_+)$,

$$f_\star(x) = \begin{cases} -1, & \text{if } \mathbb{E}[y | x] < \Delta, \\ 1, & \text{if } \mathbb{E}[y | x] > \Delta. \end{cases}$$

Exercise 3. Regression, square loss

We consider a learning problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$ and the square loss defined as $\ell(y, z) = (y - z)^2$.

1. Express the conditional risk $r(z | x)$ as a quadratic function of z .
2. Deduce the Bayes predictor $f_\star : \mathcal{X} \rightarrow \mathbb{R}$.

Solution: The conditional risk for this problem is

$$r(z | x) = \mathbb{E}[\ell(y, z) | x] = \int_{-\infty}^{\infty} (y - z)^2 p(y | x) dy.$$

Since the loss is a convex function, the conditional risk is also convex. Therefore, a minimizer is any z such that $\frac{\partial}{\partial z} r(z | x) = 0$, provided the derivative is continuous.

$$\frac{\partial}{\partial z} r(z | x) = -2 \int_{\mathbb{R}} (y - z) p(y | x) dy \stackrel{!}{=} 0 \quad (2)$$

$$\implies z \int_{\mathbb{R}} p(y | x) dy = \int_{\mathbb{R}} y p(y | x) dy. \quad (3)$$

Therefore the Bayes predictor is given by the expected value of the conditional distribution $p(\cdot | x)$:

$$f_{\star}(x) = \mathbb{E}[y | x]. \quad (4)$$

Exercise 4. Regression, absolute loss

We consider a learning problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$ and the absolute loss defined as $\ell(y, z) = |y - z|$.

1. Split the conditional risk $r(z | x)$ into integrals over $(-\infty, z]$ and $[z, \infty)$.
2. Using the first-order optimality condition, identify a Bayes predictor $f_{\star} : \mathcal{X} \rightarrow \mathbb{R}$.

Solution: The conditional risk for this problem is

$$r(z | x) = \mathbb{E}[\ell(y, z) | x] = \int_{-\infty}^{\infty} |y - z| p(y | x) dy.$$

Since the loss is a convex function, the conditional risk is also convex. Therefore, a minimizer is any z such that $\frac{\partial}{\partial z} r(z | x) = 0$, provided the derivative is continuous. Recall the definition of (conditional) cumulative distribution function (CDF):

$$C_{Y|X}(z | x) \stackrel{\text{def}}{=} \Pr(Y \leq z | X = x) = \int_{-\infty}^z p(y | x) dy = 1 - \int_z^{\infty} p(y | x) dy.$$

We compute the derivative of the conditional risk

$$\begin{aligned} \frac{\partial}{\partial z} r(z | x) &= - \int_{\mathbb{R}} \text{sign}(y - z) p(y | x) dy \\ &= + \int_{-\infty}^z p(y | x) dy - \int_z^{\infty} p(y | x) dy \\ &= C_{Y|X}(z | x) - (1 - C_{Y|X}(z | x)) = 2C_{Y|X}(z | x) - 1. \end{aligned}$$

Hence, if the CDF is continuous, the Bayes predictor (evaluated at x) is $f_{\star}(x)$ such that

$$2C_{Y|X}(f_{\star}(x) | x) - 1 = 0 \implies C_{Y|X}(f_{\star}(x) | x) = \frac{1}{2},$$

or, equivalently, $f_{\star}(x)$ is the **median** of the conditional distribution $p(y | x)$.

Exercise 5. Regression, “pinball” loss

We consider a learning problem on $\mathcal{X} \times \mathcal{Y}$, with $\mathcal{Y} = \mathbb{R}$ and the “pinball” loss defined as $\ell(y, z) = \alpha(y - z)$ if $y \geq z$ and $\ell(y, z) = (1 - \alpha)(z - y)$ if $z > y$, for $\alpha \in [0, 1]$.

1. Write the conditional risk as the sum of two integrals, separating the cases $y < z$ and $y \geq z$.
2. Compute a Bayes predictor $f_{\star} : \mathcal{X} \rightarrow \mathbb{R}$ and interpret it in terms of conditional quantiles. What happens when $\alpha = 1/2$?

Solution: The solution to this exercise is completely analogous to the previous one. We will outline the main points.

Conditional risk:

$$r(z | x) = (1 - \alpha) \int_{-\infty}^z (z - y) p(y | x) dy + \alpha \int_z^{\infty} (y - z) p(y | x) dy.$$

Derivative of the conditional risk, with respect to z :

$$\frac{\partial}{\partial z} r(z | x) = (1 - \alpha) C_{Y|X}(z | x) - \alpha (1 - C_{Y|X}(z | x)) = C_{Y|X}(z | x) - \alpha.$$

Therefore, a Bayes predictor at x is any $f_\star(x)$ such that

$$C_{Y|X}(f_\star(x) | x) = \alpha.$$

By definition, $f_\star(x)$ is a α -**quantile** of the conditional distribution $p(y | x)$, which is any number q such that $\Pr(Y \leq q | X = x) = \alpha$. Note that, for $\alpha = 1/2$ we are in the same setting of Exercise 4, recovering the same result.